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	Before increase	After increase
price	\$1,000	1.1(\$1,000) = \$1,100
earnings	\$200	1.05(\$200) = \$210
$\frac{\text{price}}{\text{earnings}}$	$\frac{1{,}000}{200} = 5$	$\frac{1{,}100}{210} = \frac{110}{21}$

Therefore, the percent increase in the ratio of price per share to earnings per share is

$$\left(\frac{\frac{110}{21}}{-} \times 100\right)\% = \left(\frac{110 - 105\,(5)}{(21)} \times 100\right)\% = \frac{500}{105}\%$$

Now substitute $k = 10$ and $m = 5$ into the answer choices to determine which answer choice gives $\frac{500}{105}\%$.

A $\frac{10}{5}\%$ NO

B (10 – 5)% NO

C $\frac{100\,(5)}{100 + 10}\% = \frac{500}{110}\%$ NO

D $\frac{100\,(5)}{100 + 5}\% = \frac{500}{105}\%$**YES**

E $\frac{100\,(5)}{100 + 10 + 5}\% = \frac{500}{115}\%$ NO

The correct answer is D.

•

Of the 300 subjects who participated in an experiment using virtual-reality therapy to reduce their fear of heights, 40 percent experienced sweaty palms, 30 percent experienced vomiting, and 75 percent experienced dizziness. If all of the subjects experienced at least one of these effects and 35 percent of the subjects experienced exactly two of these effects, how many of the subjects experienced only one of these effects?

A. 105

B. 125

C. 130

D. 180

E. 195

Arithmetic Applied Problems

Let a be the number who experienced only one of the effects, b be the number who experienced exactly two of the effects, and c be the number who experienced all three of the effects. Then $a + b + c = 300$, since each of the 300 participants experienced at least one of the effects. From the given information, $b = 105$ (35% of 300), which gives $a + 105 + c = 300$, or $a + c = 195$ (Equation 1). Also, if the number who experienced sweaty palms (40% of 300, or 120) is added to the number who experienced vomiting (30% of 300, or 90), and this sum is added to the number who experienced dizziness (75% of 300, or 225), then each participant who experienced only one of the effects is counted exactly once, each participant who experienced exactly two of the effects is counted exactly twice, and each participant who experienced all three of the effects is counted exactly 3 times. Therefore, $a + 2b + 3c = 120 + 90 + 225 = 435$. Using $b = 105$, it follows that $a + 2(105) + 3c = 435$ or $a + 3c = 225$ (Equation 2). Then, solving the system defined by Equation 1 and Equation 2,

$$\begin{cases} a + c = 195 \end{cases}$$

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$\begin{cases} a+c=195 \\ a+3c=225 \end{cases}$	multiply 1st equation by –3
$\begin{cases} -3a-3c=-585 \\ a+3c=225 \end{cases}$	add equations
$-2a=-360$, or $a=180$	

The correct answer is D.

•

If $m^{-1}=-\frac{1}{3}$, then m^{-2} is equal to

A. –9

B. –3

C. $-\frac{1}{9}$

D. $\frac{1}{9}$

E. 9

Arithmetic Negative Exponents

Using rules of exponents, $m^{-2}=m^{-1\times 2}=\left(m^{-1}\right)^2$, and since $m^{-1}=-\frac{1}{3}$, $m^{-2}=\left(-\frac{1}{3}\right)^2=\frac{1}{9}$.

The correct answer is D.

•

A photography dealer ordered 60 Model X cameras to be sold for $250 each, which represents a 20 percent markup over the dealer's initial cost for each camera. Of the cameras ordered, 6 were never sold and were returned to the manufacturer for a refund of 50 percent of the dealer's initial cost. What was the dealer's approximate profit or loss as a percent of the dealer's initial cost for the 60 cameras?

A. 7% loss

B. 13% loss

C. 7% profit

D. 13% profit

E. 15% profit

Arithmetic Percents

Given that $250 is 20% greater than a camera's initial cost, it follows that the initial cost for each camera was $\left(\$\frac{250}{1.2}\right)$. Therefore, the initial cost for the 60 cameras was $60\left(\$\frac{250}{1.2}\right)$. The total revenue is the sum of the amount obtained from selling 60 − 6 = 54 cameras for $250 each and the $\left(\frac{1}{2}\right)\left(\$\frac{250}{1.2}\right)$ refund for each of 6 cameras, or $(54)(\$250)+(6)\left(\frac{1}{2}\right)\left(\$\frac{250}{1.2}\right)$. The total profit, as a percent of the total initial cost, is $\left(\frac{(\text{ total revenue })-(\text{ total initial cost })}{(\text{ total initial cost })}\times 100\right)\%=\left(\left(\frac{(\text{ total revenue })}{(\text{ total initial cost })}-1\right)\times 100\right)\%$. Using the numerical expressions obtained above, $\frac{(\text{ total revenue })}{(\text{ total initial cost })}-1$

$(54)(250)+6\left(\frac{1}{2}\right)\left(\frac{250}{\ }\right)$

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$$= \frac{(54)(250) + 6\left(\frac{1}{2}\right)\left(\frac{250}{1.2}\right)}{(60)\left(\frac{250}{1.2}\right)} - 1 \text{ by substitution}$$

$$= \frac{54 + 3\left(\frac{1}{1.2}\right)}{(60)\left(\frac{1}{1.2}\right)} - 1 \qquad \text{by canceling 250s}$$

$$= \frac{54(1.2) + 3}{60} - 1 \qquad \text{by multiplying top and bottom by 1.2and then canceling 1.2}$$

$$= \frac{67.8}{60} - 1$$

= 1.13 – 1

=0.13

Finally, (0.13 × 100)% = 13%, which represents a profit since it is positive.

The correct answer is D.

•

Seven pieces of rope have an average (arithmetic mean) length of 68 centimeters and a median length of 84 centimeters. If the length of the longest piece of rope is 14 centimeters more than 4 times the length of the shortest piece of rope, what is the maximum possible length, in centimeters, of the longest piece of rope?

A. 82

B. 118

C. 120

D. 134

E. 152

Algebra Statistics

Let a, b, c, d, e, f, and g be the lengths, in centimeters, of the pieces of rope, listed from least to greatest. From the given information, it follows that $d = 84$ and $g = 4a + 14$. Therefore, listed from least to greatest, the lengths are, a, b, c, 84, e, f, and $4a + 14$. The maximum value of $4a + 14$ will occur when the maximum value of a is used, and this will be the case only if the shortest 3 pieces all have the same length. Therefore, listed from least to greatest, the lengths are a, a, a, 84, e, f, and $4a + 14$. The maximum value for $4a + 14$ will occur when e and f are as small as possible. Since e and f are to the right of the median, they must be at least 84 and so 84 is the least possible value for each of e and f. Therefore, listed from least to greatest, the lengths are a, a, a, 84, 84,84, and $4a + 14$. Since the average length is 68, it follows that $\frac{a + a + a + 84 + 84 + 84 + (4a + 14)}{7} = 68$, or $a = 30$.

Hence, the maximum length of the longest piece is $(4a + 14) = [4(30) + 14] = 134$ centimeters.

The correct answer is D.

•

What is the difference between the sixth and the fifth terms of the sequence 2, 4, 7, … whose n^{th} term is $n + 2^{n-1}$?

A. 2

B. 3

C. 6

D. 16

E. 17

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Algebra Simplifying Algebraic Expressions

According to the given formula, the sixth term of the sequence is $6 + 2^{6-1} = 6 + 25$ and the fifth term is $5 + 2^{5-1} = 5 + 2^4$. Then,

$$\begin{aligned}(6 + 2^5) - (5 + 2^4) &= (6 - 5) + (2^5 - 2^4)\\ &= 1 + 2^4(2-1)\\ &= 1 + 2^4\\ &= 1 + 16\\ &= 17\end{aligned}$$

The correct answer is E.

•

From the consecutive integers –10 to 10, inclusive, 20 integers are randomly chosen with repetitions allowed. What is the least possible value of the product of the 20 integers?

A. $(-10)^{20}$

B. $(-10)^{10}$

C. 0

D. $-(10)^{19}$

E. $-(10)^{20}$

Arithmetic Properties of Numbers

If −10 is chosen an odd number of times and 10 is chosen the remaining number of times (for example, choose −10 once and choose 10 nineteen times, or choose −10 three times and choose 10 seventeen times), then the product of the 20 chosen numbers will be $(-10)^{20}$. Note that $(-10)^{20}$ is less than $-(10)^{19}$, the only other negative value among the answer choices.

The correct answer is E.

•

The letters D, G, I, I, and T can be used to form 5-letter strings such as DIGIT or DGIIT. Using these letters, how many 5-letter strings can be formed in which the two occurrences of the letter I are separated by at least one other letter?

A. 12

B. 18

C. 24

D. 36

E. 48

Arithmetic Elementary Combinatorics

There are 6 ways to select the locations of the two occurrences of the letter I, and this number can be determined by listing all such ways as shown below, where the symbol * is used in place of the letters D, G, and T:

I*I**, I**I*, I***I, *I*I*, *I**I, **I*I

Alternatively, the number of ways to select the locations of the 2 occurrences of the letter I can be determined by using $\binom{5}{2} - 4 = \frac{5!\,(2!)}{(3!)} - 4 = 10 - 4 = 6$, which is the number of ways to select 2 of the 5 locations minus the 4 ways in which the 2 selected locations are adjacent.

For each of these 6 ways to select the locations of the 2 occurrences of the letter I, there are 6 ways to select the locations of the letters D, G, and T, which can be determined by using 3! = 6 or by listing all such ways:

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locations of the letters D, G, and T, which can be determined by using 3! = 6 or by listing all such ways:

DGT, DTG, GDT, GTD, TDG, TGD

It follows that the number of ways to select the locations of the 5 letters to form 5-letter strings is (6)(6) = 36.

The correct answer is D.

•

$$\frac{0.99999999}{1.0001} - \frac{0.99999991}{1.0003} =$$

A. 10^{-8}

B. $3(10^{-8})$

C. $3(10^{-4})$

D. $2(10^{-4})$

E. 10^{-4}

Arithmetic Operations on Rational Numbers

Calculations with lengthy decimals can be avoided by writing 0.99999999 as $1 - 10^{-8}$, 0.99999991 as $1 - 9(10^{-8})$ 1.0001 as $1 + 10^{-4}$, and 1.0003 as $1 + 3(10^{-4})$. Doing this gives

$$\begin{aligned}
&\frac{1-10^{-8}}{1+10^{-4}} - \frac{1-9(10^{-8})}{1+3(10^{-4})} \\
&= \frac{[1+10^{-4}][1-10^{-4}]}{1+10^{-4}} - \frac{1-9(10^{-8})}{1+3(10^{-4})} \\
&= \frac{1-10^{-4}}{1} - \frac{1-9(10^{-8})}{1+3(10^{-4})} \\
&= \frac{[1-10^{-4}][1+3(10^{-4})]-[1-9(10^{-8})]}{1+3(10^{-4})} \\
&= \frac{1+3(10^{-4})-10^{-4}-3(10^{-8})-1+9(10^{-8})}{1+3(10^{-4})} \\
&= \frac{2(10^{-4})+6(10^{-8})}{1+3(10^{-4})} \\
&= \frac{[2(10^{-4})][1+3(10^{-4})]}{1+3(10^{-4})} \\
&= 2(10^{-4})
\end{aligned}$$

The correct answer is D.

•

Last Sunday a certain store sold copies of Newspaper A for \$1.00 each and copies of Newspaper B for \$1.25 each, and the store sold no other newspapers that day. If r percent of the store's revenue from newspaper sales was from Newspaper A and if p percent of the newspapers that the store sold were copies of Newspaper A, which of the following expresses r in terms of p?

A. $\frac{100p}{125-p}$

B. $\frac{150p}{}$

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B. $\frac{300p}{250-p}$

C. $\frac{300p}{375-p}$

D. $\frac{400p}{500-p}$

E. $\frac{500p}{625-p}$

Algebra Percents

Because the number of newspapers sold at the store last Sunday is not given, assume for simplicity that number is 1,000. Also, the value of p is not given, so choose a convenient value to work with, say $p = 60$. In the table that follows, the chosen numbers are in **boldface**.

	Newspaper A	Newspaper B	Total
Percent of papers sold	$p = \mathbf{60}$	40	100
Number sold	600	400	**1,000**
Revenue	600($1) = $600	400($1.25) = $500	$1,100
Percent of revenue (r)	$\frac{600}{1{,}100} \times 100 = \frac{600}{11}$		

Now, check the answer choices using $p = 60$ to see which one yields $\frac{600}{11}$.

A. $\frac{100p}{125-p} = \frac{100\,(60)}{125-60} = \frac{6{,}000}{65}$ No (65 doesn't have 11 as a factor)

B. $\frac{150p}{250-p} = \frac{150\,(60)}{250-60} = \frac{9{,}000}{190}$ No (190 doesn't have 11 as a factor)

C. $\frac{300p}{375-p} = \frac{300\,(60)}{375-60} = \frac{18{,}000}{315}$ No (315 doesn't have 11 as a factor)

D. $\frac{400p}{500-p} = \frac{400\,(60)}{500-60} = \frac{24{,}000}{440}$ **Yes** (440 has 11 as a factor) $\frac{24{,}000}{440} = \frac{600}{11}$

E. $\frac{500p}{625-p} = \frac{500\,(60)}{625-60} = \frac{30{,}000}{565}$ No (565 doesn't have 11 as a factor)

The correct answer is D.

•

For the past n days, the average (arithmetic mean) daily production at a company was 50 units. If today's production of 90 units raises the average to 55 units per day, what is the value of n?

A. 30

B. 18

C. 10

D. 9

E. 7

Arithmetic; Algebra Statistics; Applied Problems; Simultaneous Equations

Let x be the total production of the past n days. Using the formula $\text{average} = \frac{\text{sum of values}}{\text{number of values}}$, the

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number of values

information in the problem can be expressed in the following two equations

$$50 = \frac{x}{n} \quad \text{daily average of 50 units over the past } n \text{ days}$$

$$55 = \frac{x+90}{n+1} \quad \text{increased daily averagewhen including today's 90 units}$$

Solving the first equation for x gives $x = 50n$. Then substituting $50n$ for x in the second equation gives the following that can be solved for n:

$$\begin{aligned} 55 &= \frac{50n+90}{n+1} \\ 55(n+1) &= 50n+90 && \text{multiply both sides by } (n+1) \\ 55n+55 &= 50n+90 && \text{distribute the 55} \\ 5n &= 35 && \text{subtract } 50n \text{ and 55 from both sides} \\ n &= 7 && \text{divide both sides by 5} \end{aligned}$$

The correct answer is E.

•

If a two-digit positive integer has its digits reversed, the resulting integer differs from the original by 27. By how much do the two digits differ?

A. 3

B. 4

C. 5

D. 6

E. 7

Algebra Applied Problems

Let the one two-digit integer be represented by $10t+s$, where s and t are digits, and let the other integer with the reversed digits be represented by $10s+t$. The information that the difference between the integers is 27 can be expressed in the following equation, which can be solved for the answer.

$$\begin{aligned} (10s+t)-(10t+s) &= 27 \\ 10s+t-10t-s &= 27 && \text{distribute the negative} \\ 9s-9t &= 27 && \text{combine like terms} \\ s-t &= 3 && \text{divide both sides by 9} \end{aligned}$$

Thus, it is seen that the two digits s and t differ by 3.

The correct answer is A.

•

In an electric circuit, two resistors with resistances x and y are connected in parallel. In this case, if r is the combined resistance of these two resistors, then the reciprocal of r is equal to the sum of the reciprocals of x and y. What is r in terms of x and y?

A. xy

B. $x+y$

C. $\frac{1}{x+y}$

D. $\frac{xy}{x+y}$

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E. $\frac{x+y}{xy}$

Algebra Applied Problems

Note that two numbers are reciprocals of each other if and only if their product is 1. Thus, the reciprocals of r, x, and $y$are $\frac{1}{r}, \frac{1}{x}$ and $\frac{1}{y}$, respectively. So, according to the problem, $\frac{1}{r} = \frac{1}{x} + \frac{1}{y}$. To solve this equation for r, begin by creating a common denominator on the right side by multiplying the first fraction by $\frac{y}{y}$ and the second fraction by $\frac{x}{x}$:

$\frac{1}{r} = \frac{1}{x} + \frac{1}{y}$

$\frac{1}{r} = \frac{y}{xy} + \frac{x}{xy}$

$\frac{1}{r} = \frac{x+y}{xy}$ combine the fractions on the right side

$r = \frac{xy}{x+y}$ invert the fractions on both sides

The correct answer is D.

•

Xavier, Yvonne, and Zelda each try independently to solve a problem. If their individual probabilities for success are $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{5}{8}$, respectively, what is the probability that Xavier and Yvonne, but not Zelda, will solve the problem?

A. $\frac{11}{8}$

B. $\frac{7}{8}$

C. $\frac{9}{64}$

D. $\frac{5}{64}$

E. $\frac{3}{64}$

Arithmetic Probability

Since the individuals' probabilities are independent, they can be multiplied to figure out the combined probability. The probability of Xavier's success is given as $\frac{1}{4}$, and the probability of Yvonne's success is given as $\frac{1}{2}$. Since the probability of Zelda's success is given as $\frac{5}{8}$, then the probability of her NOT solving the problem is $1 - \frac{5}{8} = \frac{3}{8}$.

Thus, the combined probability is $\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)\left(\frac{3}{8}\right) = \frac{3}{}$

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Thus, the combined probability is $\left(\frac{2}{4}\right)\left(\frac{2}{2}\right)\left(\frac{3}{8}\right) = \frac{3}{64}$.

The correct answer is E.

•

If $\frac{1}{x} - \frac{1}{x+1} = \frac{1}{x+4}$, then x could be

A. 0

B. –1

C. –2

D. –3

E. –4

Algebra Second-Degree Equations

Solve the equation for x. Begin by multiplying all the terms by $x(x+1)(x+4)$ to eliminate the denominators.

$$\frac{1}{x} - \frac{1}{x+1} = \frac{1}{x+4}$$

$$(x+1)(x+4) - x(x+4) = x(x+1)$$

$$(x+4)(x+1-x) = x(x+1) \quad \text{factor the } (x+4) \text{ out front on the left side}$$

$$(x+4)(1) = x(x+1) \quad \text{simplify}$$

$$(x+4) = x^2 + x \quad \text{distribute the } x \text{ on the right side}$$

$$4 = x^2 \quad \text{subtract } x \text{ from both sides}$$

$$\pm 2 = x \quad \text{take the square root of both sides}$$

Both -2 and 2 are square roots of 4 since $(-2^2) = 4$ and $(2^2) = 4$. Thus, x could be -2.

This problem can also be solved as follows.

Rewrite the left side as, $\frac{(x+1)}{-} = \frac{1}{x(x+1)}$, then set equal to the right side to get $\frac{1}{x(x+1)} = \frac{1}{x+4}$ Next, cross multiply: $(1)(x+4) = x(x+1)(1)$. Therefore, $x + 4 = x^2 + x$, or $x^2 = 4$, so $x = \pm 2$.

The correct answer is C.

•

$\left(\frac{1}{2}\right)^{-3}\left(\frac{1}{4}\right)^{-2}\left(\frac{1}{16}\right)^{-1} =$

A. $\left(\frac{1}{2}\right)^{-48}$

B. $\left(\frac{1}{2}\right)^{-11}$

C. $\left(\frac{1}{2}\right)^{-6}$

D. $\left(\frac{1}{8}\right)^{-11}$

E. $\left(\frac{1}{8}\right)^{-6}$

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Arithmetic Operations on Rational Numbers

It is clear from the answer choices that all three factors need to be written with a common denominator, and they thus

$$\left(\frac{1}{2}\right)^{-3} = \left(\frac{1}{2}\right)^{-3}$$

$$\left(\frac{1}{4}\right)^{-2} = \left(\left(\frac{1}{2}\right)^{2}\right)^{-2} = \left(\frac{1}{2}\right)^{-4}$$

become $\left(\frac{1}{16}\right)^{-1} = \left(\left(\frac{1}{2}\right)^{4}\right)^{-1} = \left(\frac{1}{2}\right)^{-4}$

So, $\left(\frac{1}{2}\right)^{-3}\left(\frac{1}{4}\right)^{-2}\left(\frac{1}{16}\right)^{-1} =$

$$\left(\frac{1}{2}\right)^{-3}\left(\frac{1}{2}\right)^{-4}\left(\frac{1}{2}\right)^{-4} = \left(\frac{1}{2}\right)^{-3-4-4} = \left(\frac{1}{2}\right)^{-11}.$$

The correct answer is B.

•

List T consists of 30 positive decimals, none of which is an integer, and the sum of the 30 decimals is S. The estimated sum of the 30 decimals, E, is defined as follows. Each decimal in T whose tenths digit is even is rounded up to the nearest integer, and each decimal in T whose tenths digit is odd is rounded down to the nearest integer; E is the sum of the resulting integers. If $\frac{1}{3}$ of the decimals in T have a tenths digit that is even, which of the following is a possible value of $E - S$?

I. –16

II. 6

III. 10

A. I only

B. I and II only

C. I and III only

D. II and III only

E. I, II, and III

Arithmetic Operations on Rational Numbers

First, consider $E - S$ as the total of the "contributions" made by the decimals in T with even tenths digits and the "contributions" made by the decimals in T with odd tenths digits, as shown in the following tables.

	Decimals with even tenth digit	Example
Number that occurs in T	$\frac{1}{3}\left(30\right) = 10$	
Max possible tenth digit	8	14.89999
Contribution each makes to $E - S$	Greater than 0.1	15 – 14.89999 = 0.10001
Total contribution to $E - S$	Greater than 10(0.1) = 1	
Min possible tenth digit	0	14.00001
Contribution each makes to $E - S$	Less than 1	15 – 14.00001 = 0.99999
Total contribution to $E - S$	Less than 10(1) = 10	

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Summary	1 < (evens' contribution to $E - S$) < 10	
	Decimals with odd tenth digit	Example
Number that occurs in T	$\frac{2}{3}\left(30\right) = 20$	
Max possible tenth digit	9	14.9999
Contribution each makes to $E - S$	Greater than –1	14 –14.9999 = –0.9999
Total contribution to $E - S$	Greater than 20(–1) = –20	
Min possible tenth digit	1	14.1
Contribution each makes to $E - S$	Less than or equal to –0.1	14 –14.1 = –0.1
Total contribution to $E - S$	Less than or equal to 20(–0.1) = –2	
Summary	–20 < (odds' contribution to $E - S$) ≤ –2	

Adding the summary inequalities from the tables gives

1 + (–20) < (evens' contribution to $E - S$) + (odds' contribution to $E - S$) < 10 + (–2) or $-19 < E - S < 8$.
Thus, the possible values of $E - S$ include –16 and 6, but not 10.

Note that if T contains 10 repetitions of 1.8 and 20 repetitions of 1.9, then
$S = 10(1.8) + 20(1.9) = 18 + 38 = 56$, $E = 10(2) + 20(1) = 20 + 20 = 40$, and
$E - S = 40 - 56 = -16$.

However, if T contains 10 repetitions of 1.2 and 20 repetitions of 1.1, then
$S = 10(1.2) + 20(1.1) = 12 + 22 = 34$, $E = 10(2) + 20(1) = 20 + 20 = 40$, and $E - S = 40 - 34 = 6$.

The correct answer is B.

•

If $5 - \frac{6}{x} = x$, then x has how many possible values?

A. None

B. One

C. Two

D. A finite number greater than two

E. An infinite number

Algebra Second-Degree Equations

Solve the equation to determine how many values are possible for x.

$$\begin{aligned} 5 - \frac{6}{x} &= x \\ 5x - 6 &= x^2 \\ 0 &= x^2 - 5x + 6 \\ 0 &= (x-3)(x-2) \\ x &= 3 \text{ or } 2 \end{aligned}$$

The correct answer is C.

•

Seed mixture X is 40 percent ryegrass and 60 percent bluegrass by weight; seed mixture Y is 25 percent ryegrass and 75 percent fescue. If a mixture of X and Y contains 30 percent ryegrass, what percent of the weight of the mixture is X?

A. 10%

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B. $33\frac{1}{3}\%$

C. 40%

D. 50%

E. $66\frac{2}{3}\%$

Algebra Applied Problems

Let X be the amount of seed mixture X in the final mixture, and let Y be the amount of seed mixture Y in the final mixture. The final mixture of X and Y needs to contain 30 percent ryegrass seed, so any other kinds of grass seed are irrelevant to the solution to this problem. The information about the ryegrass percentages for X, Y, and the final mixture can be expressed in the following equation and solved for X.

$0.40X + 0.25Y = 0.30(X + Y)$	
$0.40X + 0.25Y = 0.30X + 0.30Y$	distribute the 0.30 on the right side
$0.10X = 0.05Y$	subtract $0.30X$ and $0.25Y$ from both sides
$X = 0.5Y$	divide both sides by 0.10

Using this, the percent of the weight of the combined mixture $(X + Y)$ that is X is

$$\frac{X}{X+Y} = \frac{0.5Y}{0.5Y+Y} = \frac{0.5Y}{1.5Y} = \frac{0.5}{1.5} = 0.33\overline{3} = 33\frac{1}{3}\%$$

The correct answer is B.

•

How many of the integers that satisfy the inequality $\frac{(x+2)}{(x+3)} \geq 0$ are less than 5?

A. 1

B. 2

C. 3

D. 4

E. 5

Algebra Inequalities

Note: This explanation is very detailed. However, many of the steps should become more automatic with additional practice in solving nonlinear inequalities.

The expression $\frac{(x+2)}{(x+3)}$ contains three binomials, namely $x+2$, $x+3$, and $x-2$, that determine for what values of x each binomial is zero, negative, and positive. The table below summarizes these values.

Binomial	Zero for	Negative for	Positive for
$x+3$	$x=-3$	$x<-3$	$x>-3$
$x+2$	$x=-2$	$x<-2$	$x>-2$
$x-2$	$x=2$	$x<2$	$x>2$

Note that, although $x-2=0$ when $x=2$, $\frac{(x+2)(x+3)}{x-2}$ is undefined for $x=2$. However, $\frac{(x+2)(x+3)}{x-2} = 0$ when $x=-3$ and when $x=-2$. Both of these values are integers less than 5. (i)

The following number line shows the information in the table above. The vertical segments at $x=-3$, $x=-2$, and $x=2$ partition the number line into 4 intervals:

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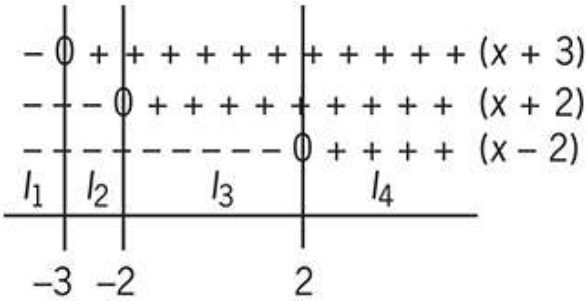
$x = 2$ partition the number line into 4 intervals:

I_1, where $x < -3$

I_2, where $-3 < x < -2$

I_3, where $-2 < x < 2$

I_4, where $x > 2$.



I_1: All three binomials are negative, so the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ is NOT satisfied.

I_2: Contains no integers, so need not be considered further.

I_3: Two binomials are positive, one is negative, so the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ is NOT satisfied.

I_4: All three binomials are positive, so the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ is satisfied. The integers in this interval that are less than 5 are 3 and 4. (ii)

From (i) and (ii), there are 4 integers, namely –3, –2, 3, and 4, for which $\frac{(x+2)(x+3)}{x-2} \geq 0$.

The correct answer is D.

•

Of the 150 houses in a certain development, 60 percent have air-conditioning, 50 percent have a sunporch, and 30 percent have a swimming pool. If 5 of the houses have all three of these amenities and 5 have none of them, how many of the houses have exactly two of these amenities?

A. 10

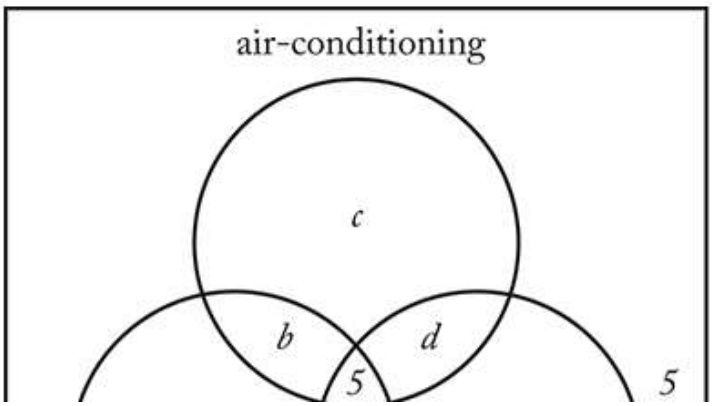
B. 45

C. 50

D. 55

E. 65

Arithmetic Sets



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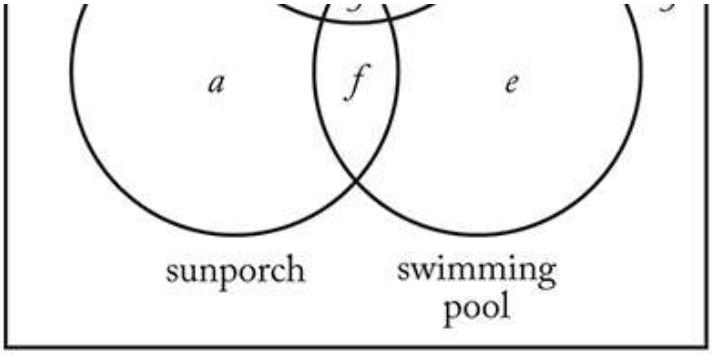

Since 60% of the 150 houses have air-conditioning, $b+c+d+5=0.6(150)=90$, so $b+c+d=85$ (i). Similarly, since 50% have a sunporch, $a+b+f+5=0.5(150)=75$, so $a+b+f=70$ (ii). Likewise, since 30% have a swimming pool, $d+e+f+5=0.3(150)=45$, so $d+e+f=40$ (iii). Adding equations (i), (ii), and (iii) gives $(b+c+d)+(a+b+f)+(d+e+f)=195$, or $a+2b+c+2d+e+2f=195$ (iv). But $a+b+c+d+e+f+5+5=150$, or $a+b+c+d+e+f=140$ (v). Subtracting equation (v) from equation (iv) gives $b+d+f=55$, so 55 houses have exactly two of the amenities.

The correct answer is D.

•

The value of $\dfrac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5}$ is how many times the value of 2^{-17}?

A. $\dfrac{3}{2}$

B. $\dfrac{5}{2}$

C. 3

D. 4

E. 5

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